

A VOXEL-BASED RAY TRACING MODEL TO SIMULATE SUNLIGHT CAPTURED BY TREES.

Raquel Melgar Fernández, January 2024.
raquel.melgar@uva.es

XVII YOUNG FOREST RESEARCHERS MEETING ON CONSERVATION AND
SUSTAINABLE USE OF FOREST SYSTEMS

OBJECTIVE OF THIS WORK

From lidar data, we want to define a metric that quantifies the amount of light that a tree captures over a certain period of time.

In our model we want to take into account:

- Relative position of the tree in the forest
- Leaf distribution on the tree crown
- Orientation of the forest
- Attenuation of light along the forest canopy

For this we will use:

- Voxelization of the data
- Beer-lambert's law
- Ray-tracing algorithms

OBJECTIVE OF THIS WORK

From lidar data, define a metric that quantifies the amount of light that a tree captures over a certain period of time.

In our model we want to take into account:

- Relative position of the tree in the forest
- Leaf distribution on the tree crown
- Orientation of the forest
- Attenuation of light along the forest canopy

1. LiDAR DATA

LiDAR: LIGHT DETECTION AND RANGING



DRON SCANNING

1. LiDAR DATA

LiDAR: LIGHT DETECTION AND RANGING



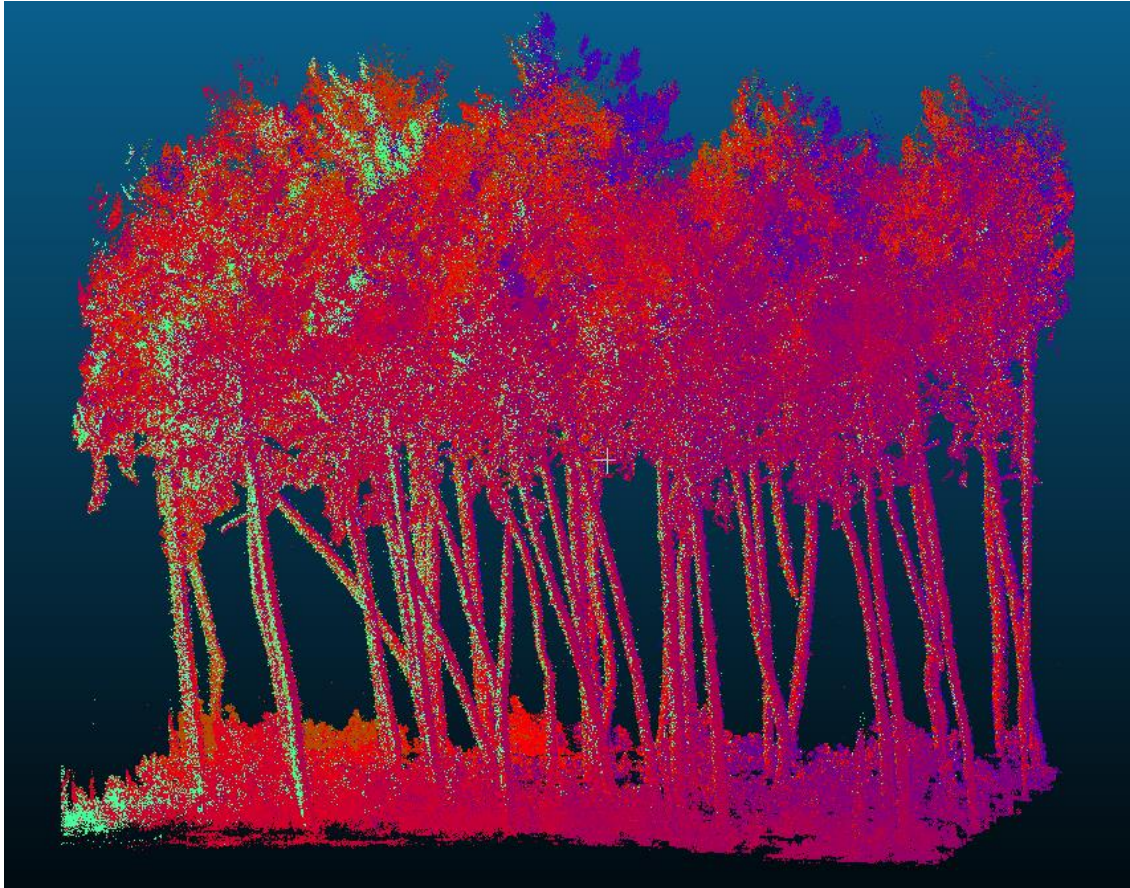
DRON SCANNING



TERRESTRIAL LASER SCANNING (TLS)

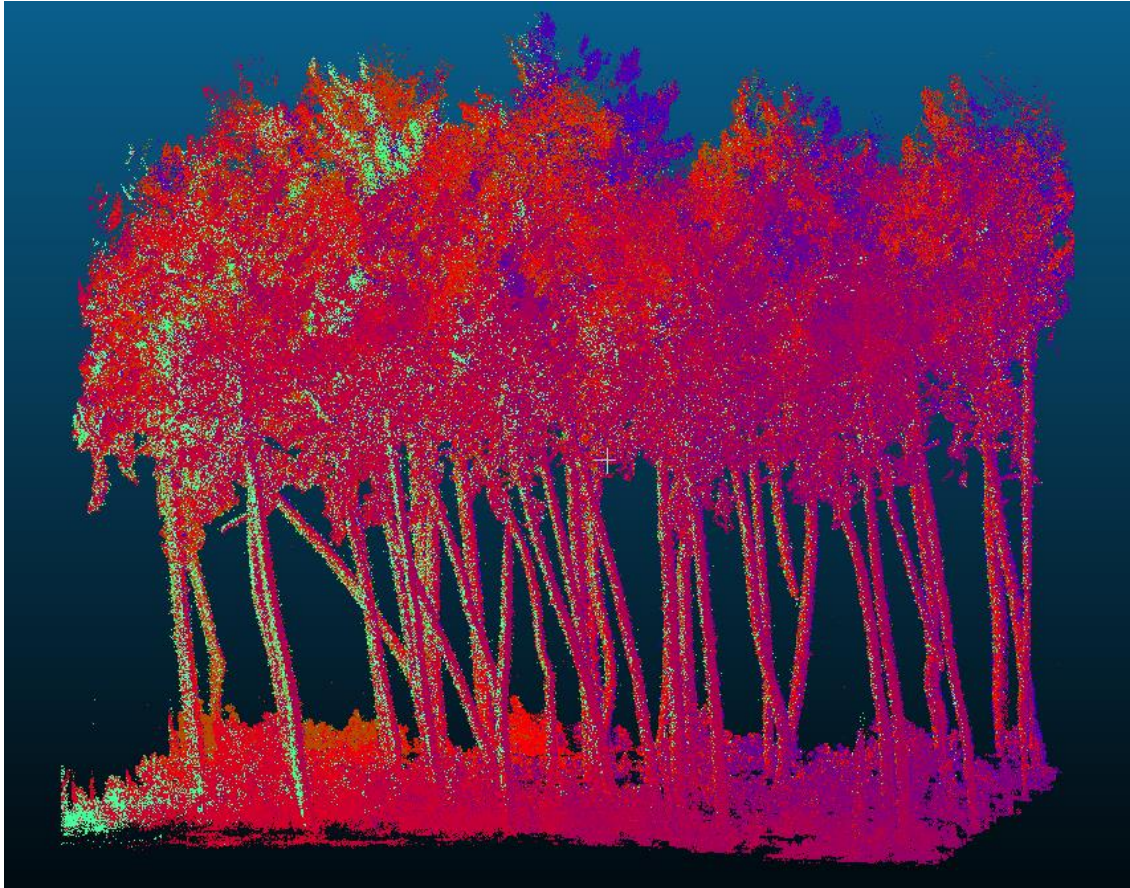
1. LiDAR DATA

OUR DATA: POINT CLOUD FROM JARANDILLA DE LA VERA
(EXTREMADURA, SPAIN)

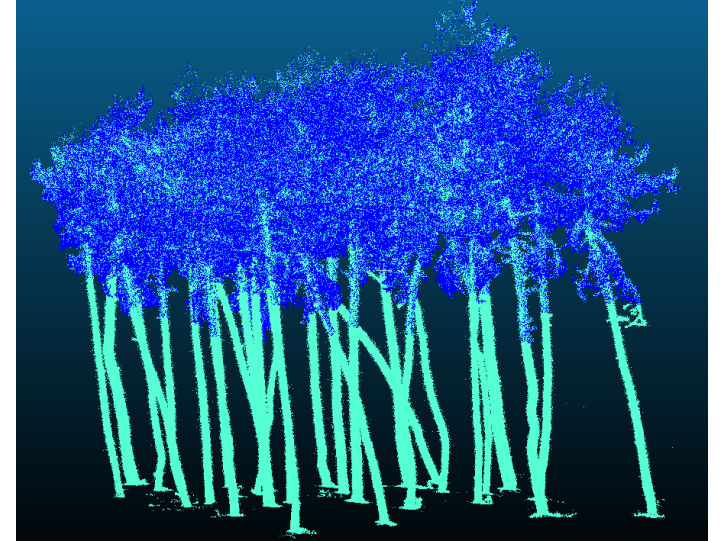


1. LiDAR DATA

OUR DATA: POINT CLOUD FROM JARANDILLA DE LA VERA
(EXTREMADURA, SPAIN)

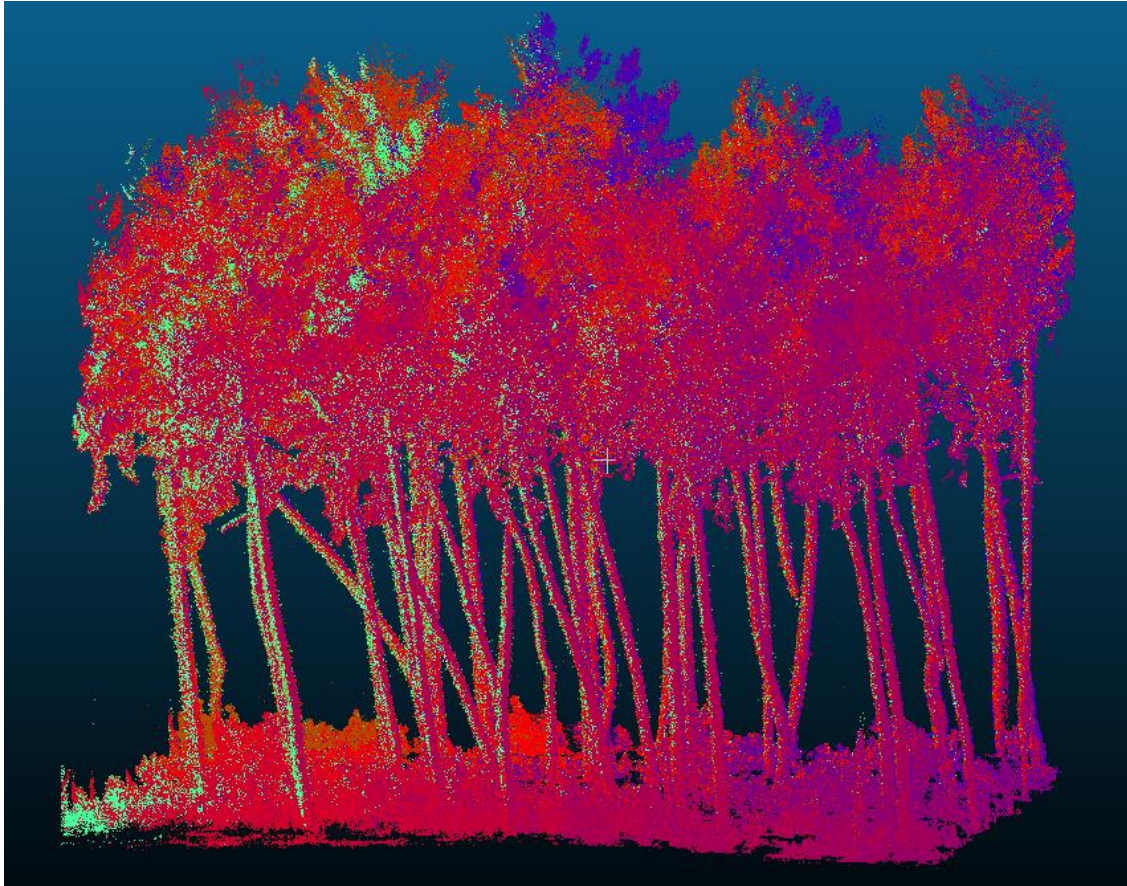


WOOD-LEAF SEGMENTATION
→

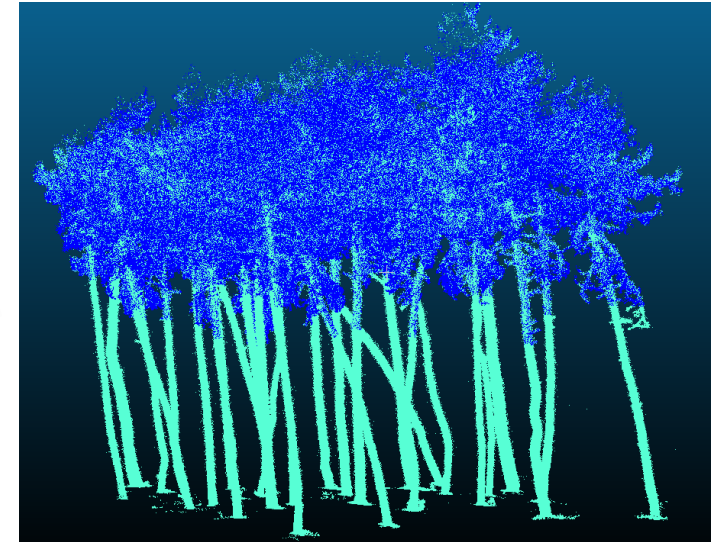


1. LiDAR DATA

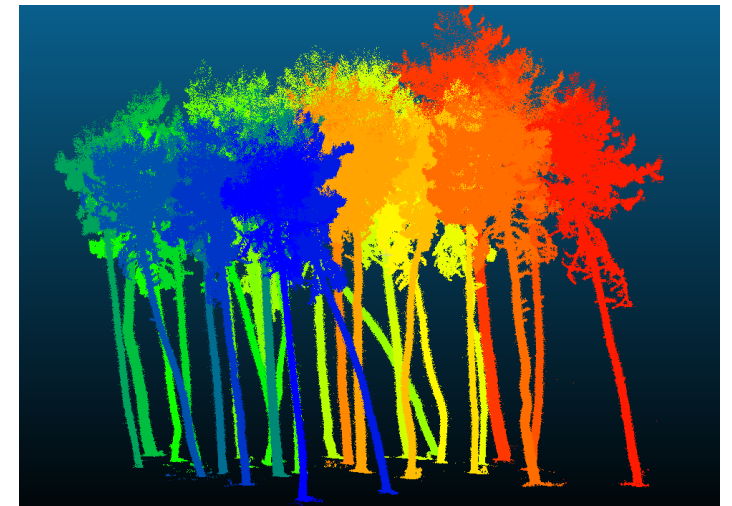
OUR DATA: POINT CLOUD FROM JARANDILLA DE LA VERA
(EXTREMADURA, SPAIN)



WOOD-LEAF SEGMENTATION



INDIVIDUAL TREE SEGMENTATION



2. WHAT ARE VOXELS?

- VOXEL (VOLUMETRIC PIXEL)
- HOW WE CAN USE VOXELS TO REPRESENT FORESTS?

2. WHAT ARE VOXELS?

- VOXEL (VOLUMETRIC PIXEL)
- HOW WE CAN USE VOXELS TO REPRESENT FORESTS?



VOXEL WIDTH: 1m

2. WHAT ARE VOXELS?

- VOXEL (VOLUMETRIC PIXEL)
- HOW WE CAN USE VOXELS TO REPRESENT FORESTS?



VOXEL WIDTH: 1m

VOXEL WIDTH: 0.7m

2. WHAT ARE VOXELS?

- VOXEL (VOLUMETRIC PIXEL)
- HOW WE CAN USE VOXELS TO REPRESENT FORESTS?



VOXEL WIDTH: 1m



VOXEL WIDTH: 0.7m



VOXEL WIDTH: 0.3m

2. WHAT ARE VOXELS?

- VOXEL (VOLUMETRIC PIXEL)
- HOW WE CAN USE VOXELS TO REPRESENT FORESTS?



VOXEL WIDTH: 1m



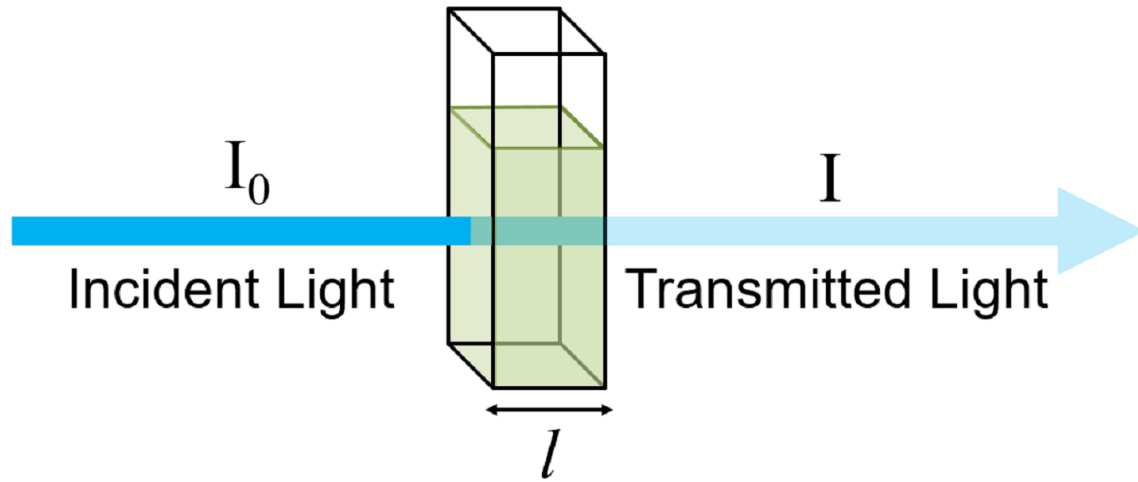
VOXEL WIDTH: 0.7m



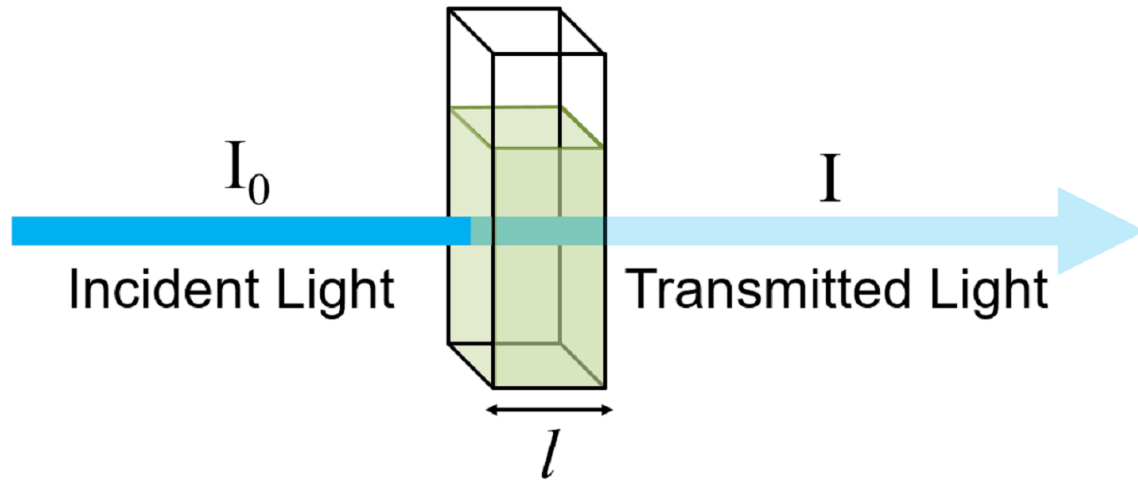
VOXEL WIDTH: 0.3m



3. SUNLIGHT EXTINCTION: BEER-LAMBERT'S LAW

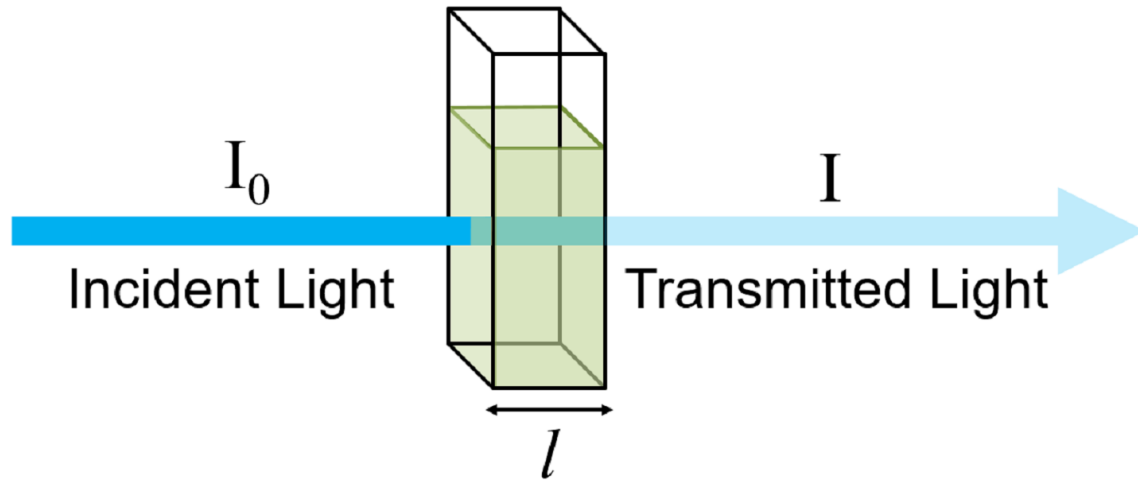


3. SUNLIGHT EXTINCTION: BEER-LAMBERT'S LAW



$$\frac{dI}{dl} = k \cdot I \implies I_s = I_e \cdot e^{-kl}$$

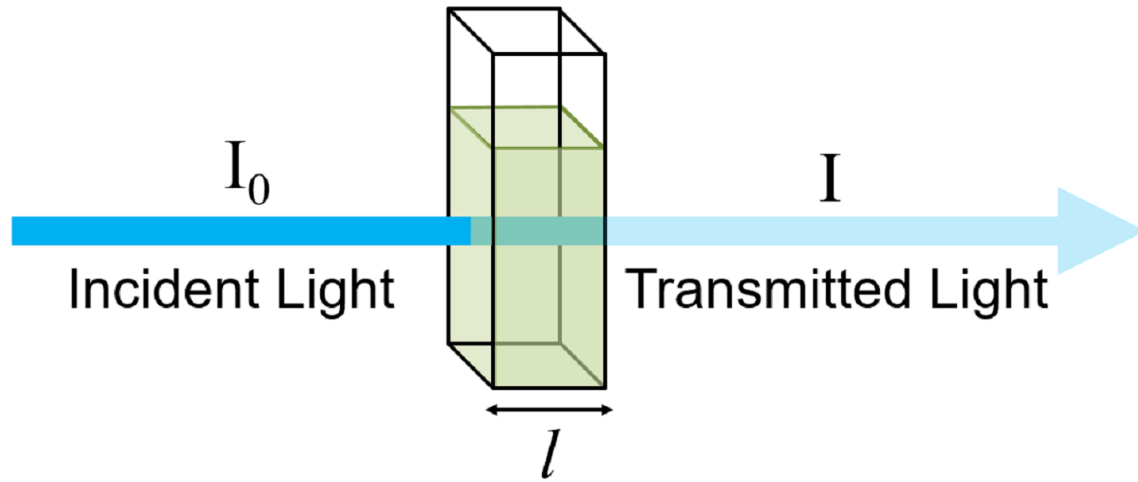
3. SUNLIGHT EXTINCTION: BEER-LAMBERT'S LAW



Beer-Lambert's law is often used to model sunlight extinction through the forest canopy

$$\frac{dI}{dl} = k \cdot I \implies I_s = I_e \cdot e^{-kl}$$

3. SUNLIGHT EXTINCTION: BEER-LAMBERT'S LAW



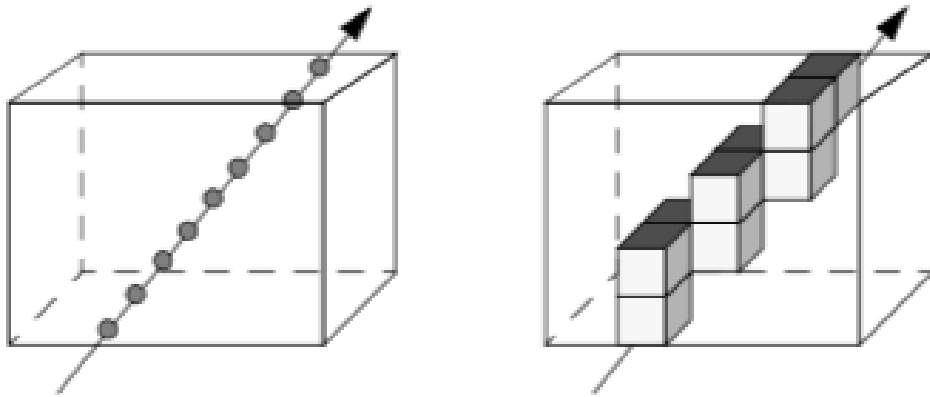
Beer-Lambert's law is often used to model sunlight extinction through the forest canopy

$$\frac{dI}{dl} = k \cdot I \implies I_s = I_e \cdot e^{-kl}$$

IDEA: ASSOCIATE A k (ATTENUATION COEFFICIENT) TO EACH VOXEL DEPENDING ON THE NUMBER OF LiDAR POINTS IT CONTAINS. SO "DARKER" VOXELS WILL ATTENUATE MORE THE LIGHT.

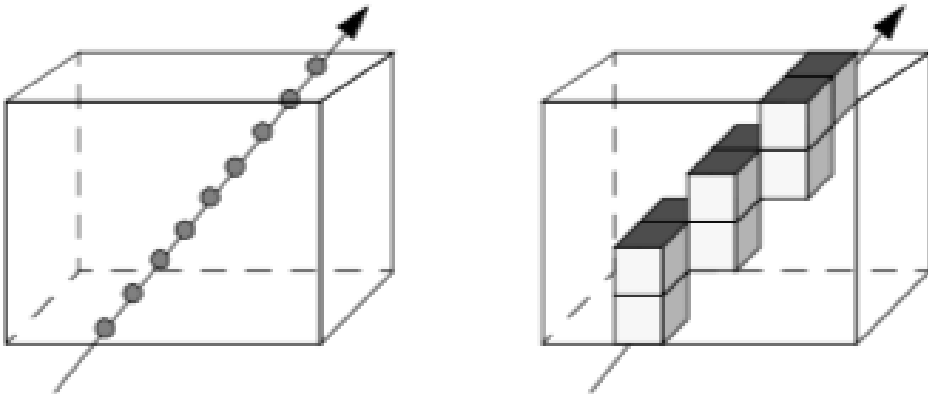
4. RAY-TRACING ALGORITHM

- Amanatides, John & Woo, Andrew. (1987). A Fast Voxel Traversal Algorithm for Ray Tracing. Proceedings of EuroGraphics. 87.

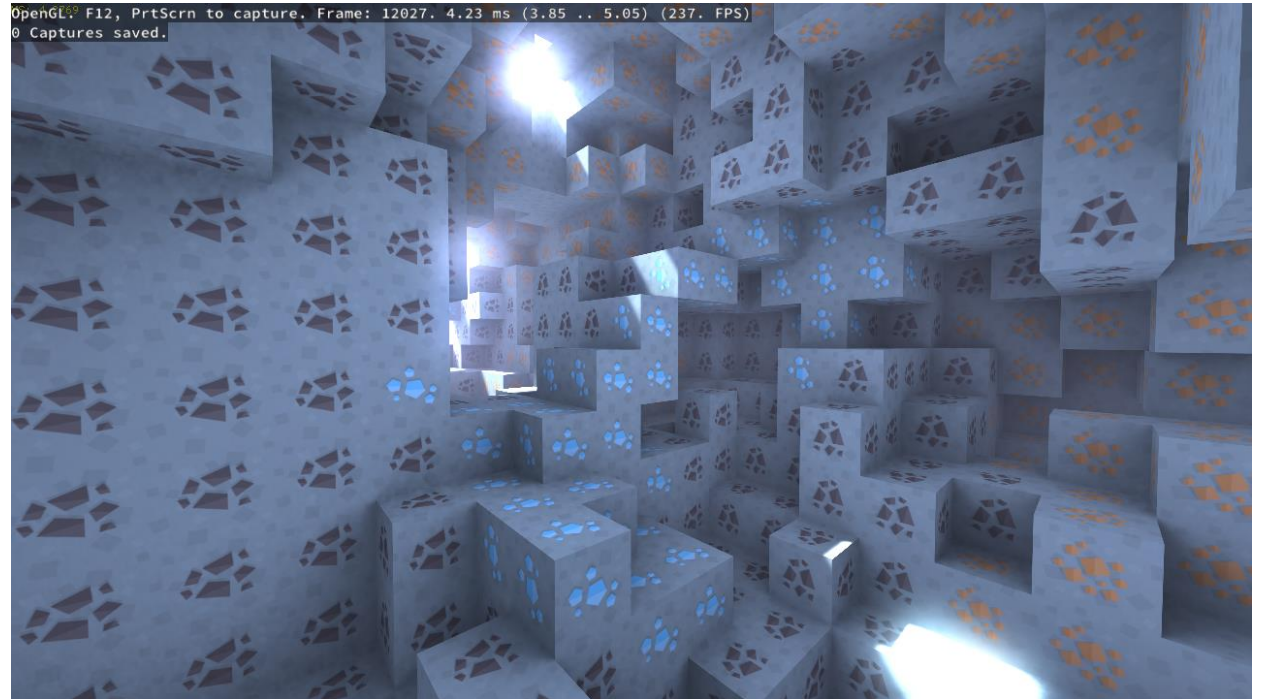


4. RAY-TRACING ALGORITHM

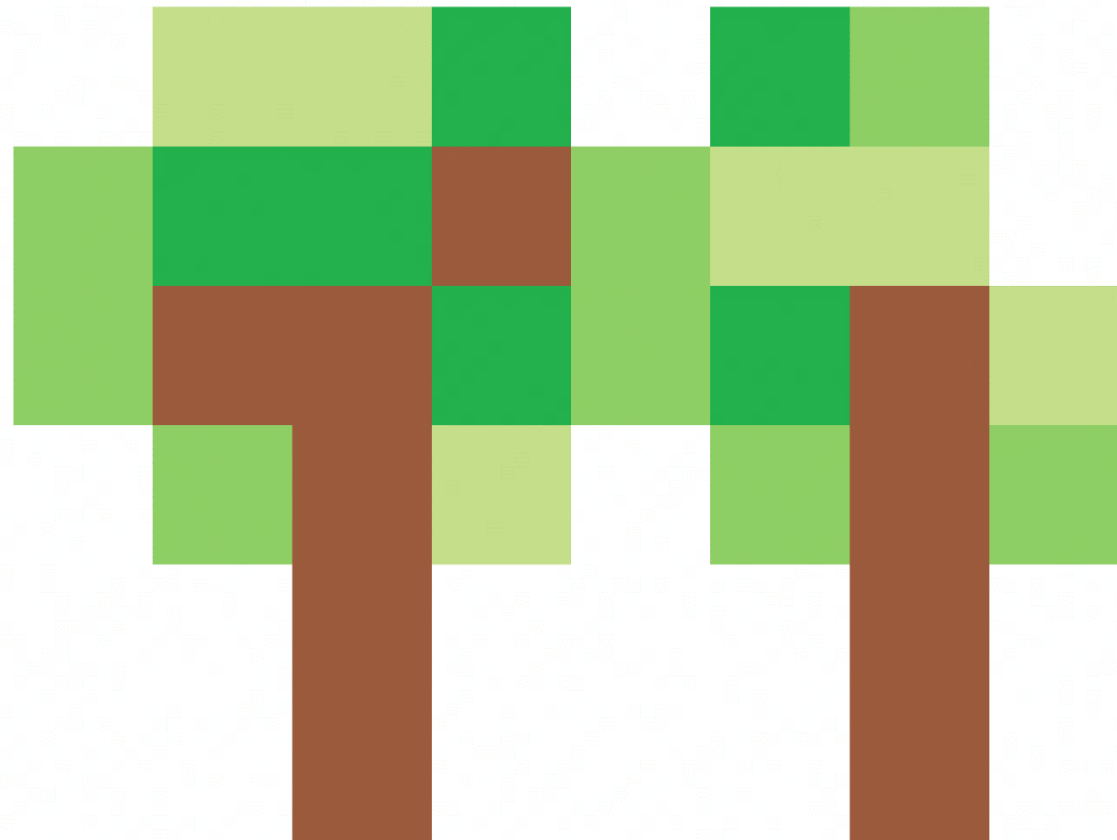
- Amanatides, John & Woo, Andrew. (1987). A Fast Voxel Traversal Algorithm for Ray Tracing. Proceedings of EuroGraphics. 87.



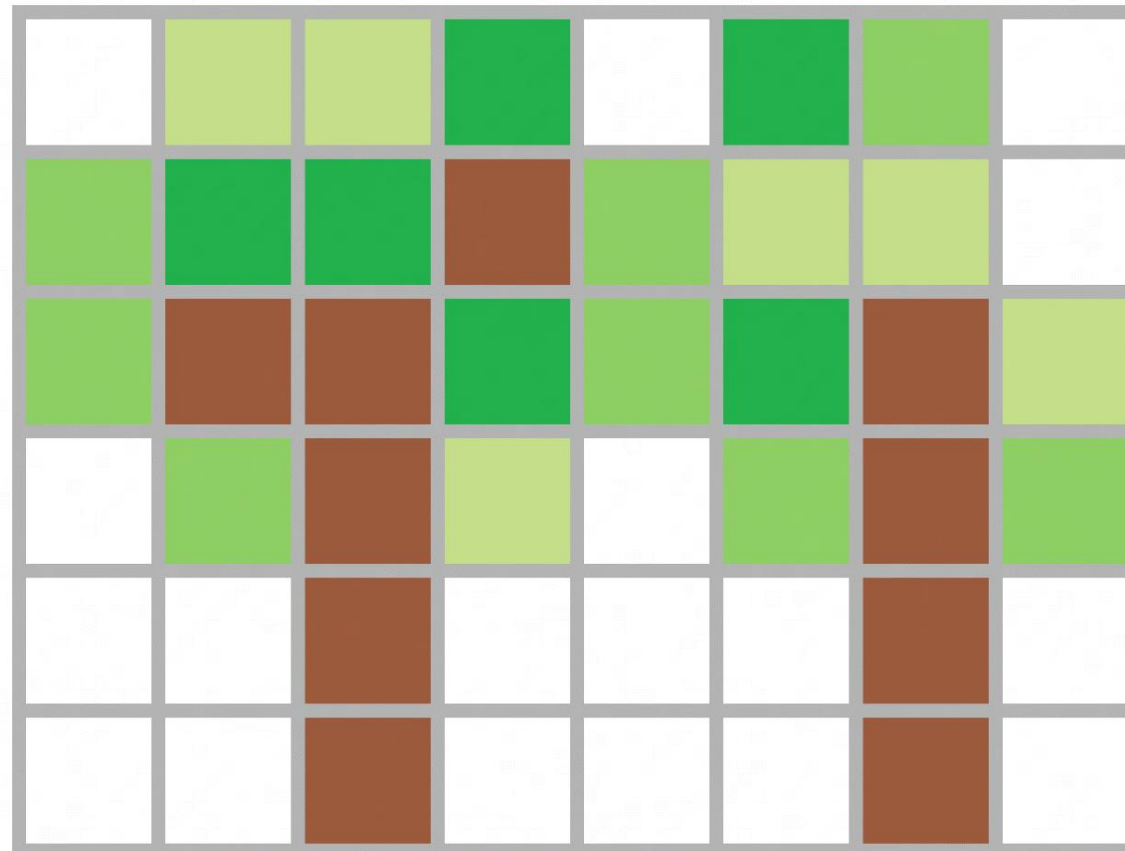
- This kind of algorithms are used in video games to simulate the interaction of light with objects and textures



5. 2D EXAMPLE OF THE ALGORITHM FOR A FIXED DIRECTION

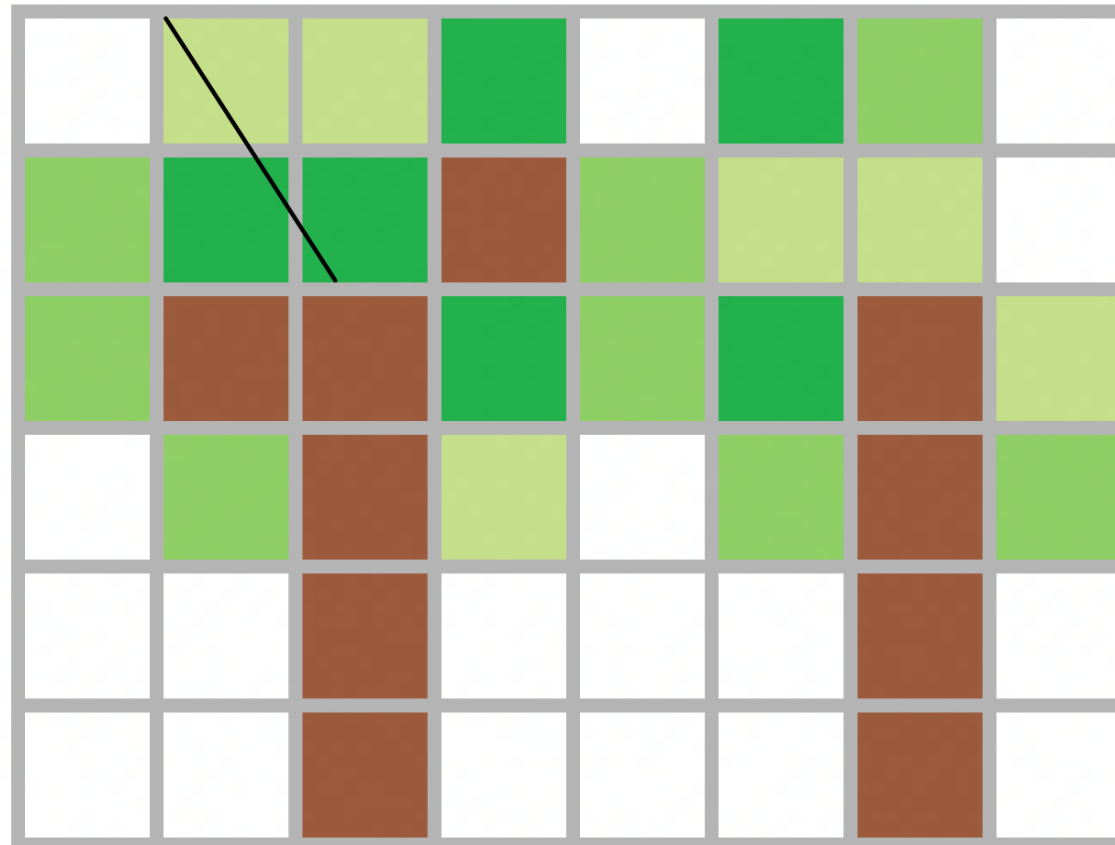


5. 2D EXAMPLE OF THE ALGORITHM FOR A FIXED DIRECTION



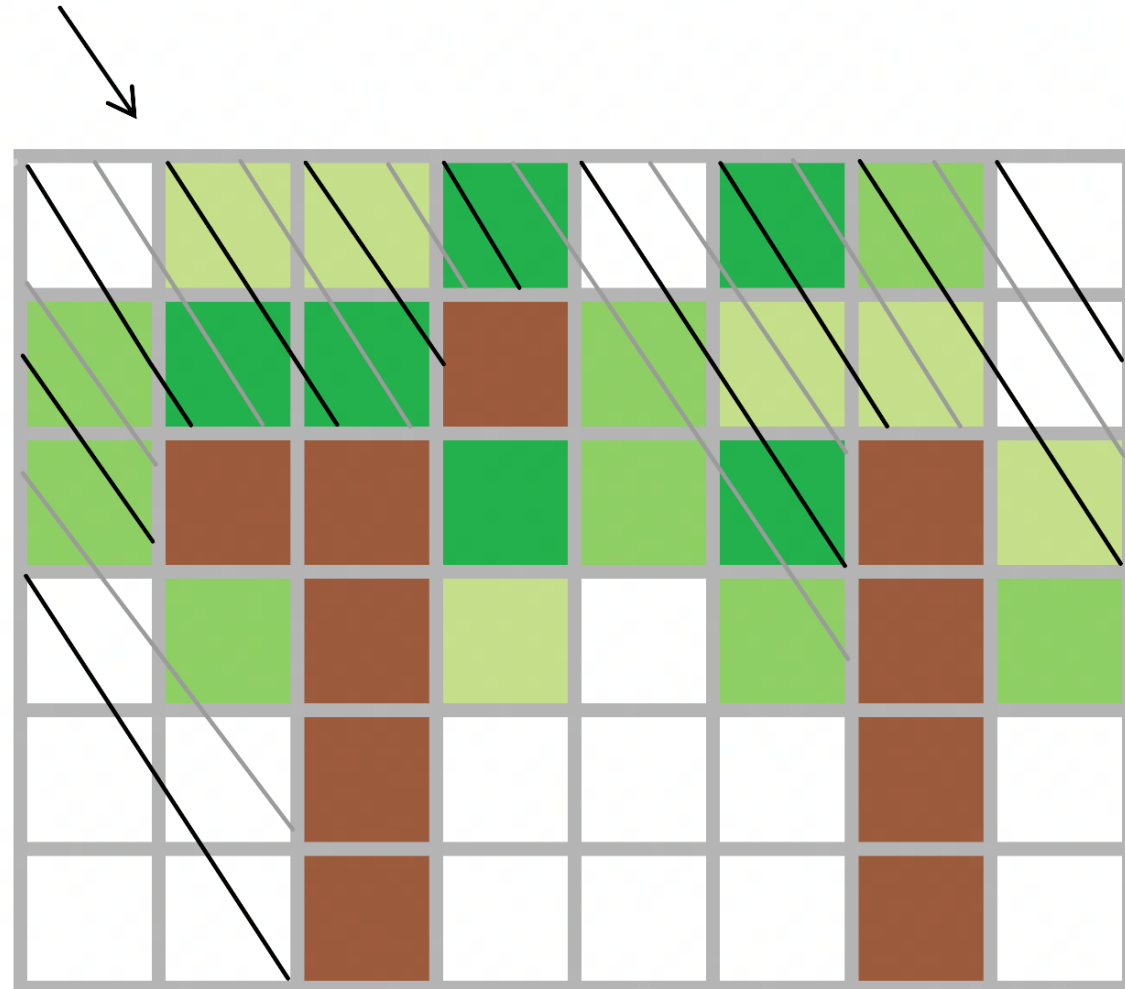
5. 2D EXAMPLE OF THE ALGORITHM FOR A FIXED DIRECTION

950 W/m²



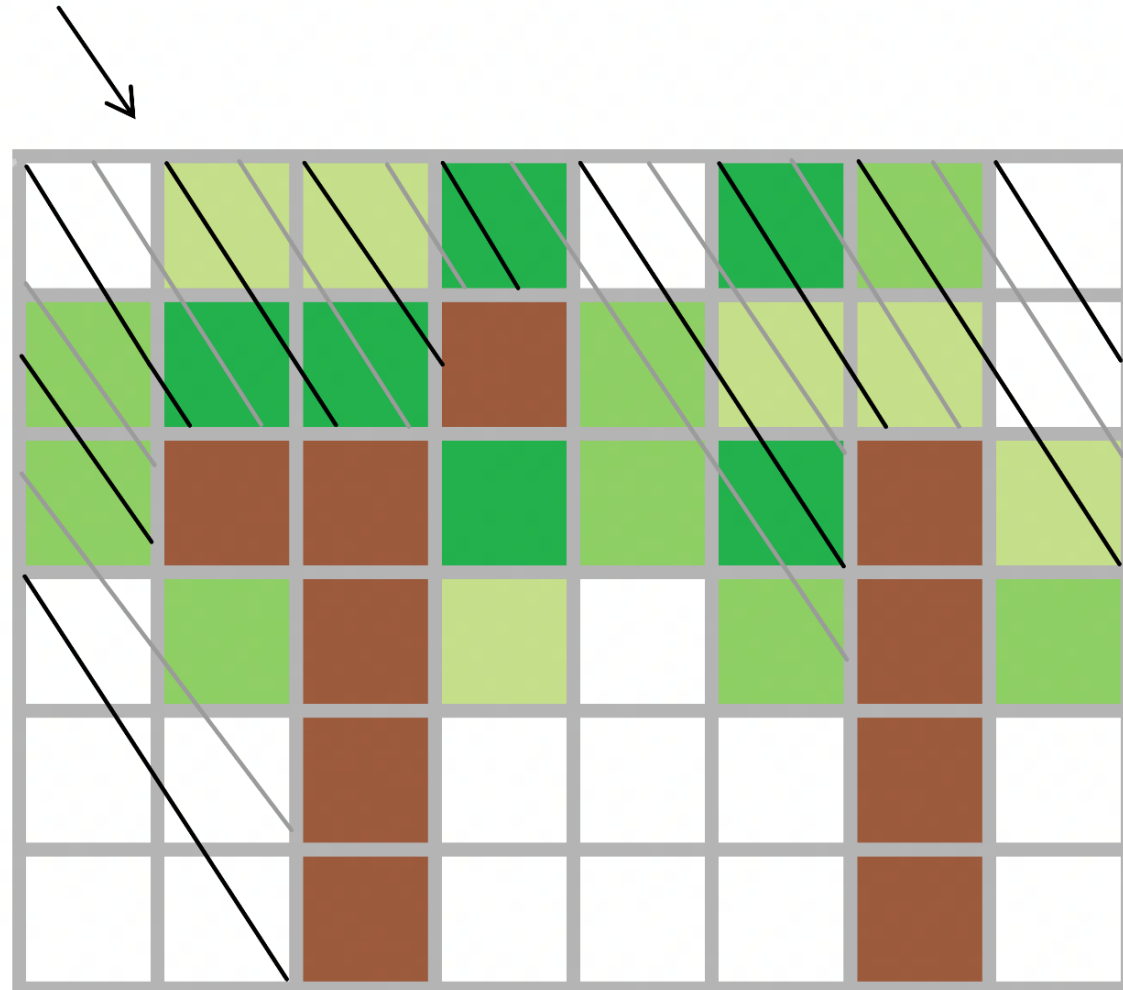
$$I_s = I_e \cdot e^{-kl}$$

5. 2D EXAMPLE OF THE ALGORITHM FOR A FIXED DIRECTION



5. 2D EXAMPLE OF THE ALGORITHM FOR A FIXED DIRECTION

We have to repeat the algorithm for each ray direction we want to simulate. The idea is to simulate the days of our tree's growing period.



6. OBSERVATIONS AND POSSIBLE IMPROVEMENTS

- The model only takes into account direct solar radiation.
- Use merged data from terrestrial and aerial scanning.
- Improve the wood-leaf separation method for the point cloud.
- Improve the K (attenuation coefficient) adjustment.
- Try to implement this on mixed forests where species have different growing periods.

THANK YOU FOR YOUR ATTENTION!!



Universidad de Valladolid

Instituto
Universitario de Investigación
iuFOR GESTIÓN
FORESTAL
SOSTENIBLE